Assignment 1: Computer Networks

Research Topic: Directed Multigraph GNNs for Real-Time AML on Cryptocurrency Transaction Networks

Phase 1: Literature Review and Research Gap Identification

Submitted by:

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Selected Papers

1. Li et al., "TcfDyG: Cryptocurrency Network Anomaly Detection Based on Time-aware Channel Fusion Dynamic Graph Neural Network," *Blockchain: Research and Applications*, 2026.
2. Lu et al., "ETX2Vec: A Fraud Detection Algorithm for Ethereum Based on Temporal Biased Random Walk Strategy," *Scientific Reports*, 2026.
3. Han et al., "Illicit Bitcoin Transaction Detection via Feature-Gated Temporal Graph Learning," *Scientific Reports*, 2026.

1. Critical Summary of Paper 1

TcfDyG: Cryptocurrency Network Anomaly Detection Based on Time-aware Channel Fusion Dynamic Graph Neural Network

Research objective and methodology

The paper proposes TcfDyG, a dynamic graph neural network for cryptocurrency anomaly detection and temporal link prediction. The model combines time-aware neighbor sampling, heterogeneous feature channels, GRU-based sequential modeling, and an MLP-Mixer-based channel fusion module. It evaluates both transductive and inductive link prediction using a chronological split: 70% training, 15% validation, and 15% testing.

Dataset and preprocessing

- Uses eight real-world temporal datasets, including Bitcoin-Alpha, Bitcoin-OTC, Fish1, Fish2, Wikipedia, Reddit, UCI, and SocialEvo.
- Bitcoin-Alpha and Bitcoin-OTC represent reputation interactions, whereas Fish1 and Fish2 contain Ethereum-related phishing transaction data.
- Historical neighbors are sampled chronologically; node, edge, and temporal information are encoded into multiple channels.
- Negative sampling is applied at a 1:1 ratio for link prediction.

Evaluation and findings

The paper evaluates Average Precision (AP), Area Under the ROC Curve (AUC), training efficiency, memory consumption, ablation performance, parameter sensitivity, and robustness for cold-start and high-frequency nodes. TcfDyG consistently performs competitively against TGAT, TGN, CAWN, DyGFormer, GraphMixer, and FreeDyG. The authors report approximately 3.9% AP and 4.9% AUC improvements on challenging phishing datasets.

Limitations and identified research gap

- **Author-reported:** dependence on high-quality labels, limited explainability, and restricted local neighborhood sampling.
- **Dataset limitation:** several datasets are interaction or reputation networks rather than complete AML transaction streams.
- **Methodological limitation:** the model is primarily designed for link prediction, not direct transaction-level AML decision-making.

- **Deployment limitation:** no complete real-time streaming pipeline, alert latency analysis, or online model-update experiment is demonstrated.
- **Research opportunity:** develop a directed multigraph AML model that preserves multiple parallel transactions, directionality, temporal order, and online inference constraints.

2. Critical Summary of Paper 2

ETX2Vec: A Fraud Detection Algorithm for Ethereum Based on Temporal Biased Random Walk Strategy

Research objective and methodology

ETX2Vec is a graph embedding approach for Ethereum fraud detection. It reconstructs local transaction subgraphs using first-order predecessor and successor neighbors and introduces four temporal biased random walk strategies. The selected TTABRW strategy uses transaction timestamps and amounts to guide the walk, after which Skip-Gram learns node embeddings for downstream node classification.

Dataset and preprocessing

- The original Ethereum graph contains approximately 2,973,489 nodes and 13,551,303 edges, with 1,165 illicit nodes.
- A subgraph is constructed using 1,165 legitimate and 1,165 fraudulent target nodes, producing 25,671 nodes and 331,945 edges.
- The graph is represented as a directed multigraph, with timestamps and transaction amounts attached to edges.
- The method extracts predecessor and successor neighbors and applies temporal non-decreasing constraints during random walks.

Evaluation and findings

The method is evaluated using accuracy, precision, recall, F1-score, and average performance. TTABRW obtains 96.04% average performance, with 95.58% precision, 96.43% recall, and 96.00% F1-score in the reported comparison. It outperforms the evaluated DeepWalk, Node2Vec, GCN, GAT, and other graph embedding baselines.

Limitations and identified research gap

- **Methodological limitation:** ETX2Vec is not a GNN-based method; it relies on random walks and Skip-Gram embeddings.
- **Sampling limitation:** only one-hop predecessor and successor neighborhoods are used, which may omit longer laundering chains.
- **Learning limitation:** the method uses a transductive and offline setting rather than continuous streaming inference.
- **Evaluation limitation:** the reported classification task is based on a constructed subgraph and may not represent a live operational AML environment.
- **Author-identified direction:** the paper suggests extending the method to other blockchains and investigating near-real-time or low-latency fraud detection through streaming or incremental updates.
- **Research opportunity:** combine its explicit directionality and transaction-amount modeling with an inductive temporal GNN for continuous AML detection.

3. Critical Summary of Paper 3

Illicit Bitcoin Transaction Detection via Feature-Gated Temporal Graph Learning

Research objective and methodology

FG-EGCN formulates illicit Bitcoin transaction detection as temporal node classification. It combines an EvolveGCN-H temporal graph encoder with a residual feature branch and a node-wise feature-gated fusion mechanism. Focal-loss-based optimization is used to improve sensitivity to rare illicit transactions under severe class imbalance.

Dataset and preprocessing

- Uses the Elliptic Bitcoin transaction benchmark, where nodes are associated with illicit, licit, or unlabeled states.
- The graph is represented as a sequence of temporal snapshots, preserving changing topology and node features.
- The temporal graph branch captures evolving relational information, while the residual branch preserves original transaction descriptors.
- Focal loss is used to address the imbalance between illicit and licit classes.

Evaluation and findings

The study reports illicit precision of 0.834, illicit recall of 0.722, illicit F1 of 0.774, and MicroAVG F1 of 0.971 for FG-EGCN in the main comparison. It also includes timestep-level analysis, ablation studies, gate-value analysis, and representation visualization. The model improves the balance between temporal graph evidence and transaction-level features.

Limitations and identified research gap

- **Dataset limitation:** the Elliptic benchmark does not fully represent address-level, entity-level, cross-chain, or suspicious-subgraph AML scenarios.
- **Methodological limitation:** the framework focuses on transaction-level node classification and does not explicitly model long-range laundering semantics or richer heterogeneous interactions.
- **Explainability limitation:** gate analysis improves representation interpretability but does not provide fully human-readable forensic explanations.
- **Deployment limitation:** online updating, low-latency inference, and operational alert processing are identified as future directions rather than fully evaluated components.
- **Research opportunity:** extend feature-gated temporal learning to directed multigraphs with edge-level transaction events and real-time risk scoring.

4. Cross-Paper Comparison Table

#	Title	Year	Dataset Used	Preprocessing Techniques	Models / Techniques Used	Evaluation Metrics	Results	Limitations / Research Gap
1	TcfDyG: Time-aware Channel Fusion Dynamic GNN	2026	Bitcoin-Alpha, Bitcoin-OTC, Fish1, Fish2, Wikipedia, Reddit, UCI, SocialEvo	Chronological 70/15/15 split; temporal historical-neighbor sampling; 1:1 negative sampling; multi-channel feature encoding	Dynamic GNN; time encoding; GRU sequential modeling; MLP-Mixer channel fusion; inductive and transductive link prediction	AP, AUC, runtime, memory, ablation, sensitivity, robustness	Approximately 3.9% AP and 4.9% AUC gains on challenging phishing datasets; competitive performance across domains	Supervised-label dependence; limited explainability; local sampling may miss global patterns; no complete real-time AML deployment evaluation
2	ETX2Vec: Temporal Biased Random Walk for Ethereum Fraud Detection	2026	Ethereum transaction graph and sampled subgraph; original graph has 2,973,489 nodes and 13,551,303 edges	Directed multigraph construction; predecessor and successor extraction; timestamp and amount-aware random walks; Skip-Gram sequences	Temporal biased random walks: TURW, TTBRW, TABRW, TTABRW; Skip-Gram node embeddings; downstream node classification	Accuracy, precision, recall, F1, average performance	TTABRW reports 96.04% average performance, 96.43% recall, and 96.00% F1	Offline transductive learning; local one-hop subgraphs; no GNN backbone; no evaluated streaming inference; limited cross-chain generalization
3	FG-EGCN: Feature-Gated Temporal Graph Learning	2026	Elliptic Bitcoin transaction benchmark with illicit, licit, and unlabeled nodes	Temporal graph snapshots; node-feature preservation; focal-loss optimization for class imbalance	EvolveGCN-H temporal encoder; residual feature projector; node-wise feature gate; temporal node classification	Illicit precision, recall, F1, MicroAVG F1, timestep analysis, ablation, gate analysis, visualization	Illicit precision 0.834; recall 0.722; F1 0.774; MicroAVG F1 0.971	Elliptic-only evaluation; limited address/entity and cross-chain modeling; no long-range laundering semantics; explanations not fully forensic; online updates left for future work

5. Research Gap Synthesis

Research Gap	Evidence from Paper 1: TcfDyG	Evidence from Paper 2: ETX2Vec	Evidence from Paper 3: FG-EGCN	Proposed Research Opportunity
Gap 1: Lack of integrated directed multigraph GNN modeling	Uses dynamic GNNs and heterogeneous channels, but the main task is link prediction and directed parallel transaction semantics are not explicitly central.	Explicitly constructs a directed multigraph and retains transaction timestamps and amounts, but does not use a GNN backbone.	Uses temporal graph learning but does not explicitly model directed multigraph edge multiplicity and detailed fund-flow direction.	Design a directed multigraph temporal GNN in which incoming and outgoing messages, repeated transactions, edge attributes, and temporal order are separately encoded.
Gap 2: Limited real-time and streaming evaluation	Chronological splits improve realism, but the experiments do not provide an end-to-end streaming alert pipeline or operational latency measurements.	The authors explicitly identify near-real-time detection and incremental embedding updates as future work.	Online updating and deployment-oriented optimization are suggested future directions rather than fully evaluated capabilities.	Develop an online inference pipeline with incremental graph updates, sliding temporal windows, latency measurement, throughput analysis, and continuous risk scoring.
Gap 3: Incomplete AML context and generalization	Several datasets are reputation or interaction networks, and the model is not specifically validated across multiple operational AML settings.	Focuses on Ethereum fraud and a constructed local subgraph, limiting broader cross-chain and entity-level conclusions.	Uses the Elliptic benchmark and does not explicitly model cross-chain behavior, entities, or long-range laundering semantics.	Evaluate across multiple cryptocurrency datasets and investigate address-level, entity-level, and subgraph-level AML detection, including cross-chain or heterogeneous relations where data is available.
Gap 4: Explainability for financial investigation	The authors identify the hybrid architecture as a black box and call for post-hoc explanations.	Embeddings and random-walk paths offer structural information but do not provide analyst-oriented forensic explanations.	Gate analysis improves representation understanding but does not produce complete human-readable explanations.	Add explainable risk evidence such as influential transactions, suspicious paths, temporal motifs, edge contributions, and counterfactual explanations for investigators.